

EDUCATION

Carnegie Mellon University Expected May 2027

M.S. in Software Engineering | GPA: 4.0 / 4.0

Selected Coursework: Distributed Systems, Foundations of Software Engineering, Software Refactoring

University of Wisconsin–Madison May 2024

B.S. in Computer Science & B.S. in Computer Engineering | GPA: 3.72 / 4.0

Selected Coursework: Operating Systems, Computer Architecture, Data Structure & Programming III, Algorithms

WORK EXPERIENCE

NewsBreak May – August 2026

Software Engineering Intern Mountain View, CA

- Redesigned a cache that could not hit (13.2MB keyed per user) after pod memory neared its 512MiB limit; re-keyed it to a shared 10.9MB catalog (78× smaller) and cut the home-load copy 9.0ms → 0.89ms.
- Replaced a 25–30 request client fan-out with one backend call under a 3-second per-source budget, and paged a 206-topic catalog so the first page fell from 2.8MB (~4.7s) to ~0.27MB.
- Built a FastAPI / MongoDB backend from the first commit to 89 endpoints with ~1,450 fully-offline tests (0.17s) behind a ruff / pyright / pytest gate; a Motor fake reproduces a production failure as a regression guard.
- Designed a news pipeline that shortlists near-duplicates by embedding cosine, then runs an LLM same-story dedupe so a user never gets two alerts for one event; calibrated over 4,100+ decisions and cut filler 34% → 4%.
- Diagnosed a 5-day silent CronJob outage (non-terminal ImagePullBackOff + Forbid concurrency skipping every later run) and fixed it with paired job and scheduler deadlines.
- Root-caused a 65% failure rate in another team’s LLM workers: one stalled read killed an `asyncio.gather` batch and an unbounded loop ran to 16,384 tokens; sized a 2,048-token cap.
- Built hybrid search (substring + text index + vector, fused by RRF) where any failing leg degrades to the survivors; deterministic entity resolution took real queries from 0% to 100% P@1 at 0.4ms per resolve.
- Sole SwiftUI engineer and primary backend author for [SportsBreak](#) (~6,000 installs, 1,119 peak daily active users); authored 344 pull requests across five repositories (340 merged).

Haddee Education June – September 2025

Engineering Lead San Diego, CA

- Led an 8-intern team through Agile sprints to ship a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; designed its database schemas and REST APIs, led PR reviews, and mentored interns on Git workflows and testing protocols.

ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch September – December 2023

Full Stack Developer Beijing, China

- Built a VC-backed startup’s LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; demoed it until three 200–500-person companies signed contracts.

SKILLS

|                   |                                                                                     |
|-------------------|-------------------------------------------------------------------------------------|
| Languages         | C, Python, Go, Java, SQL, Bash, TypeScript, Swift                                   |
| Systems & Testing | Linux, asyncio, Redis Streams, Unit Testing, TDD, pytest, ruff, pyright, CI/CD, Git |
| Data & Backend    | MongoDB, PostgreSQL, MySQL, FastAPI, REST APIs, hybrid retrieval, RRF               |
| Cloud & Infra     | AWS, GCP, Kubernetes (EKS), Docker, Helm, CronJobs, Jenkins, GitHub Actions         |
| Certifications    | AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner     |

PROJECTS & LEADERSHIP

- codingplans:** Benchmarking paid LLM coding plans (Z.AI, MiniMax, Kimi) with KV-cache, long-context, and needle-in-a-haystack tests to show which providers nerfed caching or heavily quantized their models.
- Wisconsin Autonomous:** Integrated wheel-speed encoders with voltage-to-frequency converters and streamed real-time data to the onboard NVIDIA Jetson.
- ECE Leadership & Engagement Fellow** (CMU, Aug 2026 – present): Own budgeting, marketing, and outreach for ECE student programs — company-sponsored hackathons, alumni events, and tech event outings.